INTERNSHIP REPORT

MAKING A PROTOTYPE OF SUPERCAPACITORS AND THEIR TESTING

Submitted by

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Indian Institute of Technology (IIT) Madras** for providing me the valuable opportunity to carry out my internship in the field of **supercapacitors**. This internship has been an enriching learning experience and has greatly contributed to my academic growth.

I extend my heartfelt thanks to **Prof. Tiju Thomas**, Department of Metallurgical and Materials Engineering, for permitting me to work in his esteemed research group. I am deeply grateful to his research scholar, **Ms. Sindhu**, for her constant guidance, technical support, and encouragement throughout the internship. Her patience, mentoring, and insightful suggestions played a crucial role in helping me understand experimental techniques and research methodologies related to supercapacitor fabrication and characterization.

I would also like to express my sincere thanks to **Dr. Ilangovan**, Head of the Department, University of Madras, for recommending me for this internship and for his continuous support and motivation. His encouragement enabled me to pursue this valuable opportunity and enhance my practical knowledge in the field of materials science.

Finally, I thank all the research scholars and staff members of the lab for their friendly support and for creating a positive and productive learning environment. This internship has been a significant milestone in my academic journey, and I am truly grateful for the experience.

RM Priyadharshini
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ABSTRACT

This study focuses on the development and electrochemical evaluation of biomass-derived activated carbon for supercapacitor applications. Bamboo powder was used as the primary carbon precursor and subjected to KOH activation, hydrothermal treatment, carbonization, and subsequent purification to obtain highly porous activated carbon with enhanced surface area. The synthesized material was incorporated into nickel foam electrodes using a PVDF-based slurry and evaluated in both two-electrode and three-electrode configurations. Electrochemical characterization was performed using Cyclic Voltammetry (CV), Galvanostatic Charge–Discharge (GCD), and Electrochemical Impedance Spectroscopy (EIS) to analyse capacitive behaviour, charge–storage capability, and internal resistance. The two-electrode device demonstrated a specific capacitance of 43.33 F g^{-1}, while the three-electrode setup yielded 26.66 F g^{-1}, indicating the material's favourable performance under both practical and intrinsic testing conditions. EIS analysis revealed low series resistance and distinct charge-transfer characteristics, confirming efficient ion diffusion and good electrode conductivity. The results highlight the potential of sustainable, low-cost bamboo-derived activated carbon as a promising electrode material for high-performance, eco-friendly supercapacitors. This work demonstrates an effective approach to converting agricultural waste into value-added energy-storage materials, supporting future advancements in green energy technologies.

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INTRODUCTION

I undertook my internship at IIT Madras, focusing on the research and development of supercapacitor electrode materials; internship was hosted in the Department of Metallurgical and Materials Engineering at IIT Madras under the mentorship of Prof. Tiju Thomas, a leading researcher in sustainable energy storage solutions. The project focused on the development of eco-friendly supercapacitor electrodes produced from paddy waste and other bio-waste, contributing to sustainable energy solutions for India, with potential applications in energy storage and electric mobility and green energy technologies and rural empowerment. The aim was to explore novel carbon-based electrodes made from biomass.

A **supercapacitor**, also known as an electrochemical capacitor or ultracapacitor, is an advanced energy-storage device that bridges the gap between conventional capacitors and batteries. While traditional capacitors can deliver high power but store very little energy, and batteries can store high energy but deliver low power, supercapacitors combine the advantages of both. They possess **high power density**, **fast charge–discharge capability**, and an **extremely long cycle life**, making them suitable for applications such as backup power, electric vehicles, and electronic devices.

Supercapacitors store energy through either **electric double-layer formation** or **fast surface redox reactions**. Based on this mechanism, they are broadly classified into three types: **Electric Double Layer Capacitors (EDLCs)**, which use carbon-based materials and store charge by ion adsorption; **pseudo capacitors**, which use metal oxides or conducting polymers and store energy through surface redox reactions; and **hybrid capacitors**, which combine both mechanisms to improve performance. Their simple structure, high stability, and quick response make them an important component in modern energy technology.

The fabrication and testing of supercapacitors generally involve two different systems: the **two-electrode system** and the **three-electrode system**. Each system serves a distinct purpose and provides different types of information about the device and the material.

The **two-electrode system** represents the actual, practical configuration of a working supercapacitor. This assembly mimics the real device and is used to evaluate performance parameters such as specific capacitance, energy density, power density, and cycle stability. In a typical two-electrode design, both electrodes are prepared using an active material—commonly activated carbon—mixed with a conductive additive and a binder. This mixture is made into a slurry and coated onto a current collector such as aluminium foil. After drying and pressing, two electrodes of equal size are prepared. These electrodes are soaked in an electrolyte and separated by a porous separator to prevent short-circuiting. Finally, they are assembled in a cell casing such as a coin cell, pouch cell, or Swagelok cell. This configuration provides realistic device-level performance and is essential for evaluating practical applications.

In contrast, the **three-electrode system** is primarily used for fundamental **electrochemical characterization** of the electrode material. It helps in understanding the intrinsic electrochemical behaviour without the influence of the second electrode, which often causes potential drop or resistance in the two-electrode system. The three-electrode setup consists of a **Working Electrode (WE)**, **Counter Electrode (CE)**, and **Reference Electrode (RE)**. The working electrode contains the active material coated on a substrate such as nickel foam, FTO glass, graphite, or metal foil. The counter electrode, usually platinum or graphite, balances the current flow, while the reference electrode, such as Ag/AgCl or Calomel, provides a stable and known reference potential. All three electrodes are immersed in an electrolyte, and the system is connected to a potentiostat for electrochemical testing. Techniques such as Cyclic Voltammetry (CV), Galvanostatic Charge–Discharge (GCD), and Electrochemical Impedance Spectroscopy (EIS) are used to analyse the material's capacitance, redox behaviour, rate capability, and charge-transfer resistance.

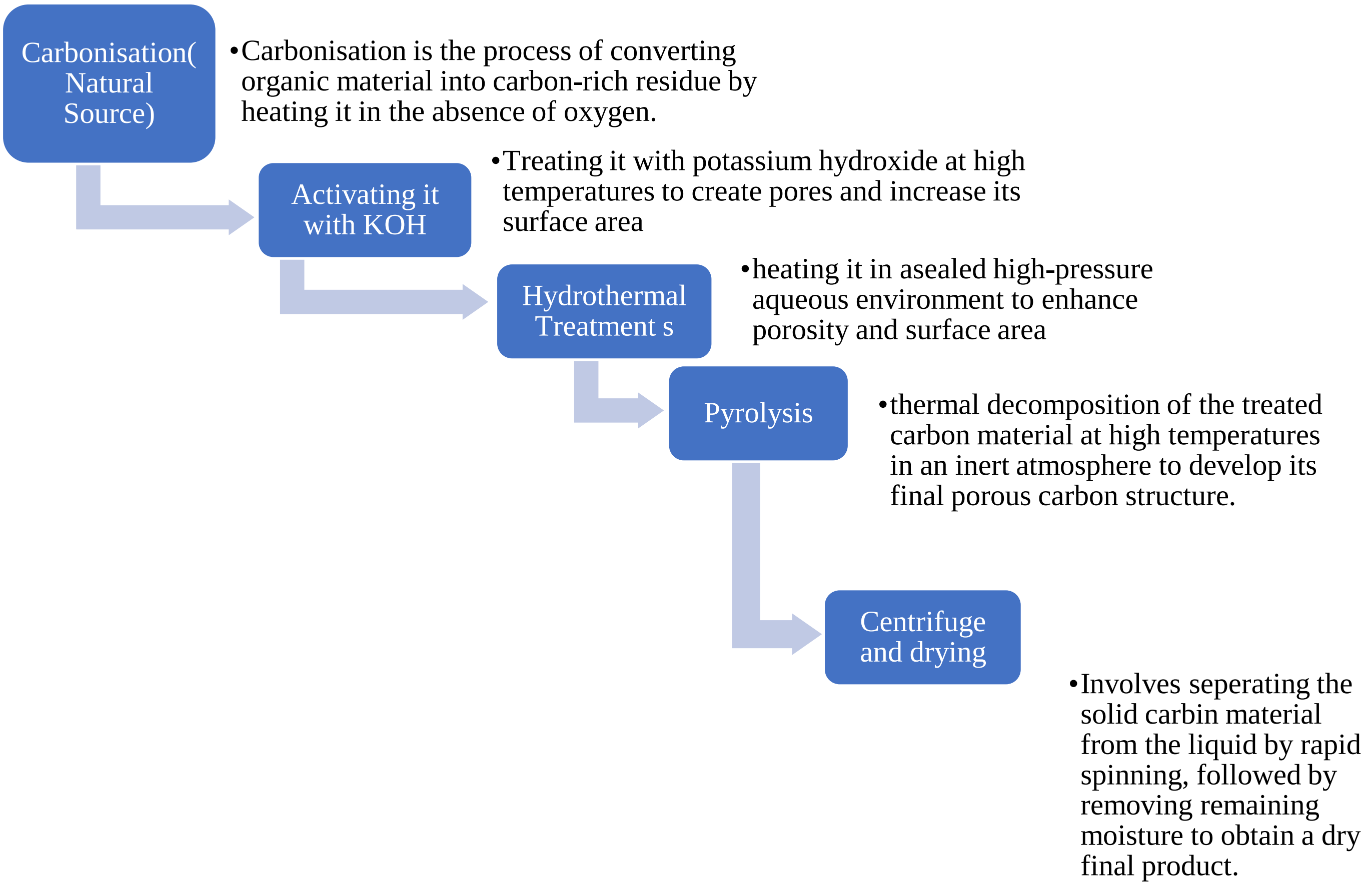
The main difference between the two systems lies in their purpose and accuracy. The two-electrode system provides realistic device-level results but generally shows lower measured capacitance due to the internal resistance and symmetric configuration. The three-electrode system, on the other hand, provides more accurate and higher capacitance values since the potential of the working electrode is precisely controlled by the reference electrode. Thus, the two-electrode system is essential for evaluating practical performance, while the three-electrode system is ideal for studying fundamental electrochemical properties.

In conclusion, supercapacitors are promising energy-storage devices due to their fast kinetics, long lifespan, and high-power capabilities. Understanding the fabrication and characterization through both two-electrode and three-electrode systems is crucial for developing high-performance supercapacitor materials. Together, these systems provide comprehensive insight into both material behaviour and real-device efficiency, enabling the design of next-generation energy-storage technologies.

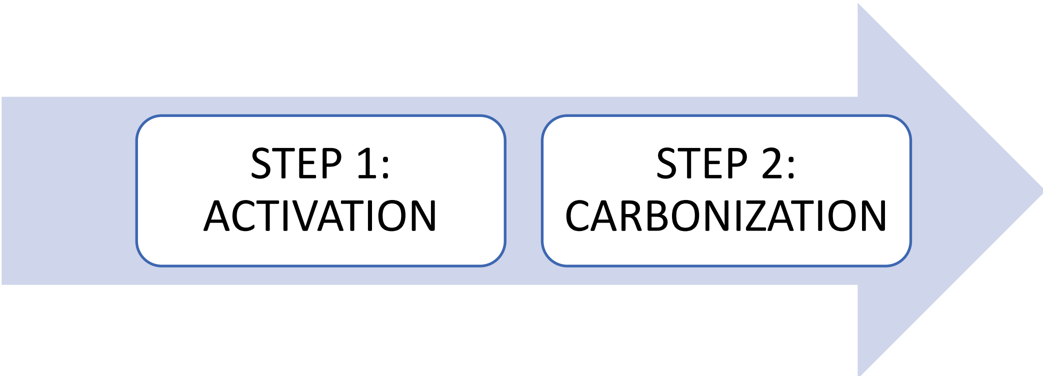
MAKING OF A SUPERCAPACITOR

SYNTHESIS OF AN ACTIVATED CARBON

METHOD 1:



METHOD 2:



STEP 1: ACTIVATION

❖ Add bamboo powder into KOH solution in the volume of 25 ML. Such that (Bamboo powder: KOH::1:2).
❖ Stir for 8 hours at room temperature.

❖ Take out the beaker from stirring and remove the magnetic beads and keep it for drying in oven for overnight at 90°C.
❖ It will form solid substance.

STEP 2: CARBONIZATION

❖ Now subject it to tubular furnace passing inert N_2 gas for two hours at 800°C.
❖ Here we used Alumina crucible/Boat.
❖ Then centrifuge with 6 wt.% HCl and DI water(de-ionised water).
❖ Keep it for drying overnight at 90°C.

And u will obtain an Activated Carbon.

MAKING OF ELECTROLYTE

For 3 Electrode System

❖ Preparation of KOH Solution: Electrolyte preparation for KOH in a 3-electrode supercapacitor involves dissolving the required concentration of KOH (typically 1M or 6M) in distilled water, stirring until fully dissolved, and using the solution as the electrolyte for electrochemical testing.

For 2 Electrode System

❖ Electrolyte preparation of KOH for a 2-electrode supercapacitor involves dissolving the desired concentration of KOH (commonly 1M–6M) in distilled water, stirring until completely dissolved, and then soaking the electrodes and separator with this solution during cell assembly.

CLEANING OF NICKEL FOAM

1. Take a Ni foam sheet.
2. It should be cut into pieces such that the coated area of Ni-Co Mof/Ac is of similar size to that of platinum foil.
3. For 2 Electrode System: We cut based on Swagelok cell size in a circular shape using an Electrode cutter.

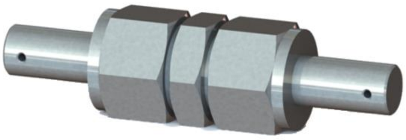


Figure 3 Swagelok cell

Figure 1 Electrode cutter

Figure 2 Nickel foam sheet

For 3 Electrode System: We cut 1.5 cm × 1 cm

Figure 4 (A)for 3 Electrode (B)for 2 Electrode

4. Cut up to as many pieces of Ni foam.

5. These Ni foam pieces are then kept inside a beaker (500 mL)

6. Sonication

Figure 5 Sonicator

I. **HCl wash**:

(8.3 mL (37%) HCl + 100 mL DI water → Liquid solution)

Now in this liquid solution add those Ni foam strips and keep it in the sonicator for 10 minutes. Now drain off the liquid after 10 minutes right after the sonicator goes off.

II. **Acetone wash:**
Take 50 mL of Acetone or enough acetone to fully immerse the available Ni foam strips. Again keep it in the sonicator for another 5 minutes then drain off the acetone into the sink.

III. **DI water wash:**
Now take DI water pour it on to the Ni foam strips after treating with Acid wash and Acetone wash such that the Ni foam strips are fully submerged. Now keep it in the sonicator for 5 minutes. Now drain off the liquid into sink.

7. Now take out those Ni foam strips and keep it in a petri dish in such a way that they do not over lie on each other separating the 2 Electrode and 3 Electrode strips separately in each petridish.

8. Now keep it in the Oven

both are ideal can be kept based on time requiremnts

For 12 to 16 hours at 60°C

For 6 hours at 80°C is enough and ideal.

Here we kept the Ni foam in petridish and that is kept in the oven at 56°C for 12 to 16 hours.

9. And coming next day or after 12 to 16 hours the Oven is switched off and the petri dish is taken out carefully using a heat resistant gloves(Auto clave gloves).

10. Now taken out nickel foam is stored inside a zipped poly ethylene bag covered by a butter sheet for further use and storage.

11. Here two electrode and Three electrode Ni foam strips are stored separately.

COATING OF SYNTHESIZED MATERIAL IN CLEANED NI FOAM

Steps involved in coating of Active material in Ni foam by brush coating method

1. Now Ni foam cleaned and dried is kept aside safely in vacuum plastic bag is covered by a butter sheet.
2. Now arrange the cleaned Ni foam strips in a petridish.
3. Take or measure each Ni foam bare weight using a weighing machine.
4. Mark the areas in petridish.
5. In order to coat, a slurry is prepared which here acts as an active material

Activated carbon (any type) + Conducting Carbon (Carbon Black) + Binder (PVDF) = Active Material (Slurry)

6. The above-mentioned materials are taken in such a way that keeping in mind how much Ni strips to be coated (here no. of Ni strips going to be coated is 6 no's)

∴6 × 2mg = 12mg ≈ 15mg of slurry needs to be prepared.

7. Now in order to mix all these three an NMP(N-Methyl-2-pyrolidone) is added.

Figure 6 weighing machine

8. Preparation of slurry

i. Take Ni-Co Mof/Ac 85% that is about 12.75mg.
ii. Then take carbon powder (conducting carbon) of about 10% that is approximately 1.5mg.
iii. Then take PVDF 5% that is about 0.75mg.
iv. Now after measuring them in weighing machine and getting the required amount mix them all together in a mortar using a pestle with addition of 100mL of NMP using a micropipette to form a slurry.

Figure 7 Things required to form a slurry

9. Now using paint brush coat the slurry such that 80% of the Ni foam is coated with slurry (active material).
10. Now keep it in oven for 6 hours at 80℃ and calculate its mass after taking it out of the oven using heat resistant gloves.

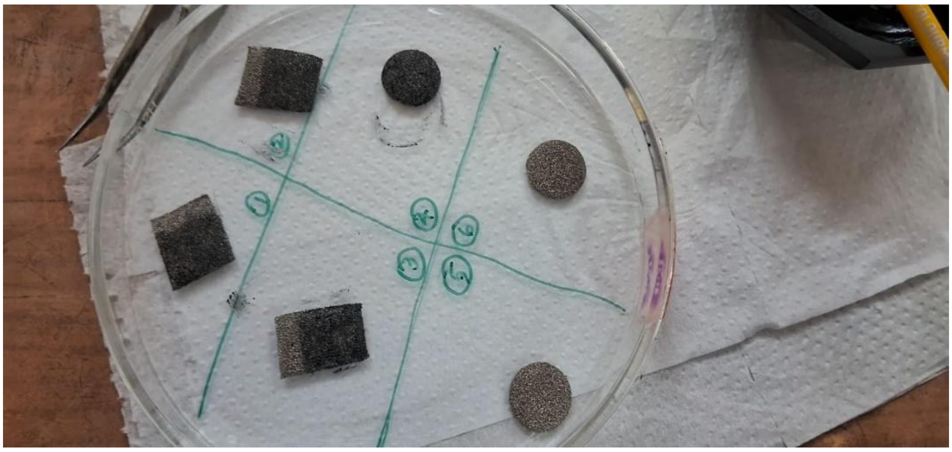


Figure 8 80% brush coated Ni foam strips with Active material

EXPERIMENTAL SETUP

For 2 Electrode System

The 2-electrode system uses two electrodes: a **positive electrode** and a **negative electrode**, assembled face-to-face with a separator soaked in electrolyte. Both electrodes act together as working and counter electrodes, mimicking a real supercapacitor device. The assembled cell can be placed in a coin cell, Swagelok cell, or held between conductive clips. The potentiostat is connected to the two terminals only (no reference electrode). This setup is used to measure practical device performance, including capacitance, energy density, and operating voltage window.

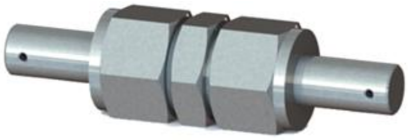


Figure 9 Swagelok cell for 2 Electrode system

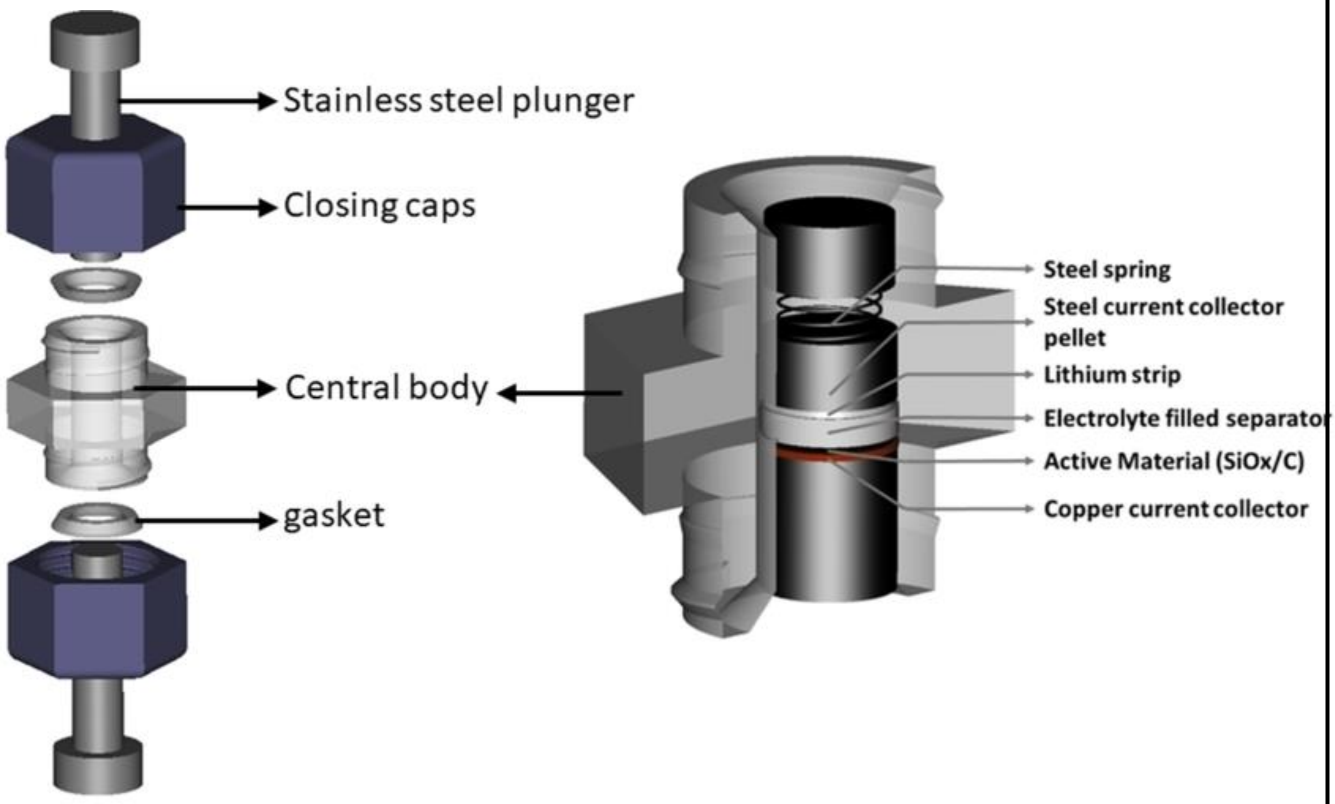


Figure 10 Inside of a Swagelok cell

For 3 Electrode System

The 3-electrode setup consists of a **working electrode (WE)** coated with the active material, a **reference electrode (RE)** such as Hg/HgO SCE(standard calomel Electrode), and a **counter electrode (CE)** platinum is used here. The working electrode is the one being studied, while the reference electrode provides a stable, fixed potential for accurate voltage measurement. The counter electrode completes the circuit and carries the current. All three electrodes are immersed in the electrolyte inside a beaker, and they are connected to the potentiostat for CV, GCD, or EIS measurements. This setup allows precise characterization of the material without interference from the opposite electrode.

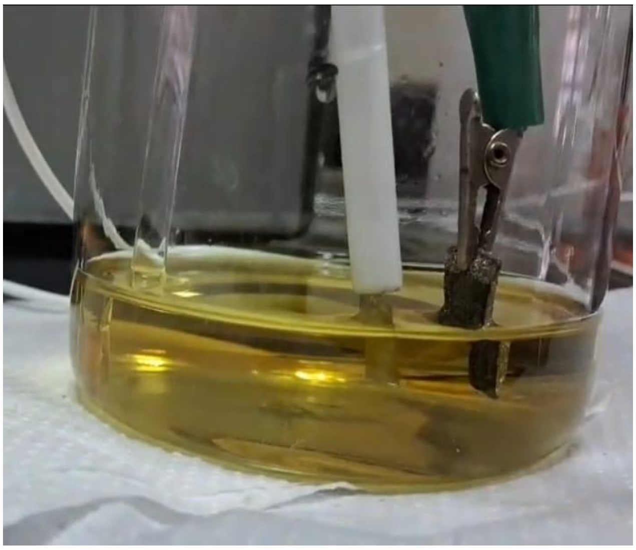
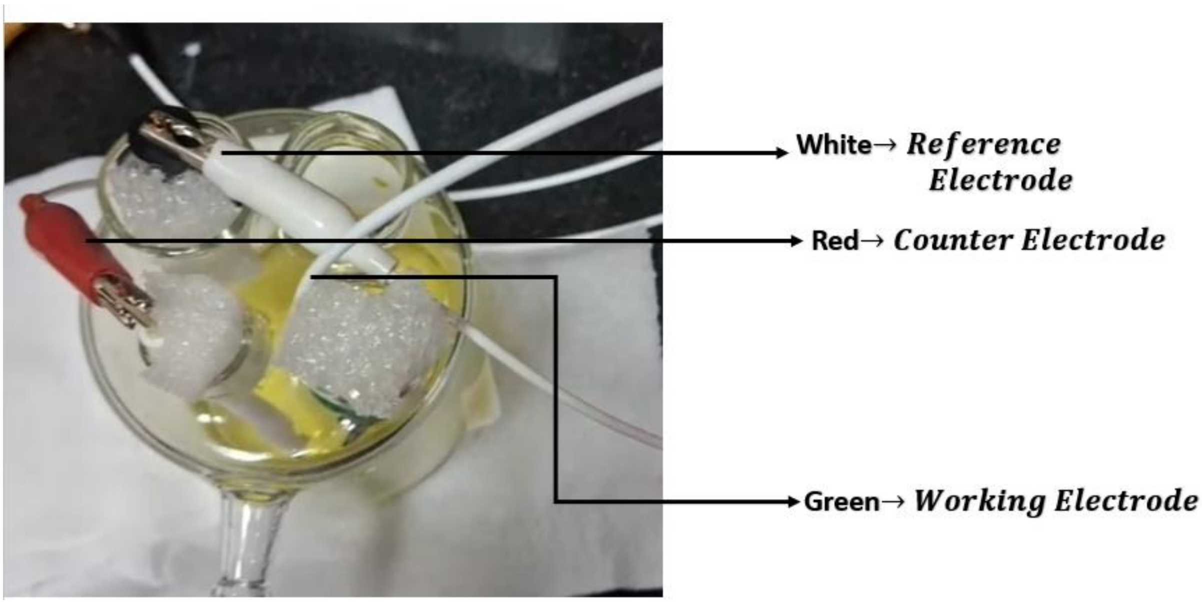


Figure 11 Three Electrode experimental set up consisting of working electrode counter electrode and reference electrode



CHARACTERIZATION

Electrochemical performance assessed by cyclic voltammetry(CV),Galvanostatic charge-discharge(GCD),and Electrochemical Impedance Spectroscopy(EIS)

For 2 Electrode System

Cyclic Voltammetry

Cyclic voltammetry shows **how easily a material gains or loses electrons** by scanning the voltage forward and backward and recording the current.

Cyclic voltammetry (CV) is used because it gives **quick, clear information about the electrochemical behaviour of a material or system**. It’s one of the most important techniques in electrochemistry and energy-storage research (supercapacitors, batteries, sensors, etc.).

Cyclic voltammetry in a two-electrode system is performed to evaluate the practical performance of the fully assembled supercapacitor device. Since both electrodes act together, the CV curve represents real working conditions of the device. This setup helps determine device-level capacitance, charge–storage behaviour, and operational voltage window. It also indicates whether the device behaves like an EDLC, pseudo capacitor, or hybrid system.

Why we need cyclic voltammetry

1. **To study redox reactions**
   CV shows when oxidation and reduction happen by producing characteristic peaks.

2. **To know reversibility**
   It tells whether a reaction is reversible, quasi-reversible, or irreversible.

3. **To understand electrochemical mechanisms**
   Helps identify reaction steps, intermediates, and electron-transfer processes.

4. **To check electrochemical stability window**
   Especially important for electrolytes in supercapacitors/batteries.

5. **To evaluate electrode performance**
   Shape of the CV curve indicates capacitive behaviour, battery-type behaviour, or mixed behaviour.

6. **To calculate specific capacitance**
   Integral area under the CV curve helps estimate capacitance for supercapacitors.

7. **To study diffusion properties**
   Peak current vs. scan rate gives information on whether the process is surface-controlled or diffusion-controlled.

8. **Quick and non-destructive**
CV is fast and gives a lot of information without damaging the electrode.

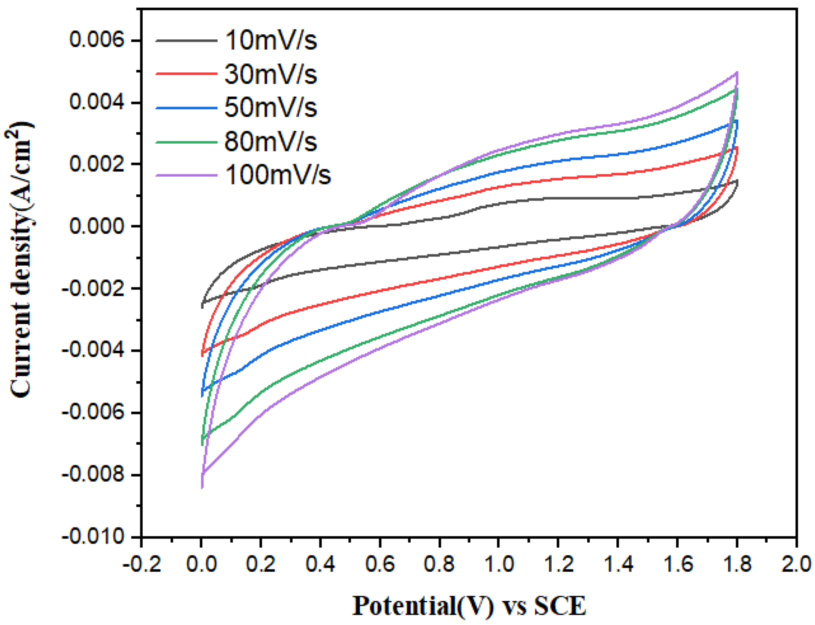


Figure 12 CV graph for 2 electrode system

Calculation of Specific Resistance from CV curve:

$$C_s = \frac{Polygon\ area\ from\ Origin}{2V scan\ rate \times \triangle\ V \times mass\ loaded}$$

$$Units = \frac{Coulombs\ (area)}{2 \times \frac{mV}{s} \times V \times g}$$

Here,

Scan rate=30 $\frac{mV}{s}$ Area=0.0039 g=0.003g

$$C_s = \frac{0.0039}{2 \times 30\frac{mV}{s} \times 0.5V \times 0.003g}$$

$$C_s = \frac{0.0024}{2 \times (30 \times 10^3)\frac{mV}{s} \times 0.5V \times 0.003g}$$

$$C_s = \frac{0.0039}{2 \times 30 \times 0.5 \times 0.003}\left[\frac{C.s}{g.v^2}\right]$$

$$C_s = \frac{0.0039}{0.00009}\left[\frac{C.s}{g.v^2}\right]$$

$$\because \left[\frac{C.s}{v^2}\right] = F(Farad)$$

$$C_s = \left(\frac{0.0039}{0.00009}\right)\frac{F}{g}$$

$$C_s = 43.33\frac{F}{g}$$

$$\therefore\ specific\ Capacitance\ from\ Cv\ curve\ is\ calculated\ to\ be\ 43.33\frac{F}{g}$$

Galvanostatic charge discharge(GCD)

GCD is needed to measure the real charge–storage performance, stability, and internal resistance of a supercapacitor under constant current.

GCD testing was conducted on the assembled 2-electrode supercapacitor to evaluate its practical energy-storage performance. The near-triangular charge–discharge curves confirmed capacitive behaviour, while the discharge time and IR drop were used to determine capacitance, energy density, and internal resistance of the device.

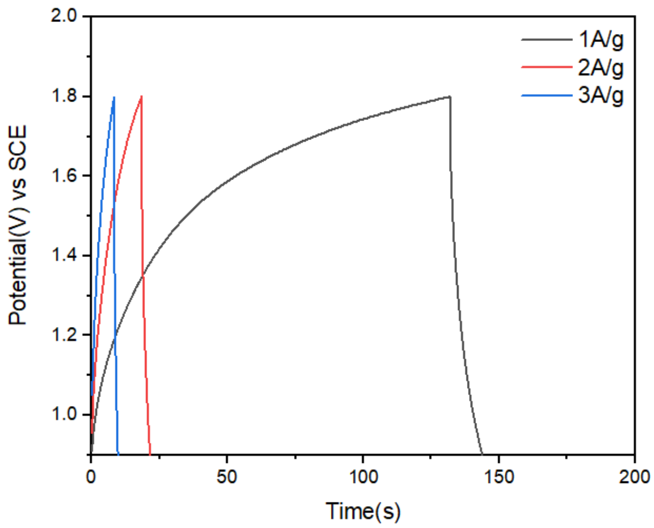
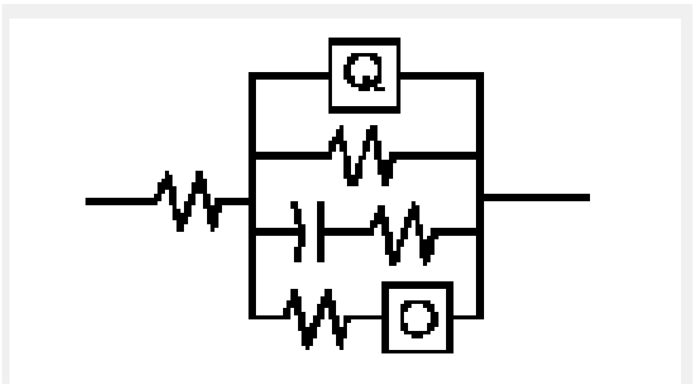


Figure 13 GCD graph for 2 Electrode system

EIS

Circuit for the following Nyquist plot



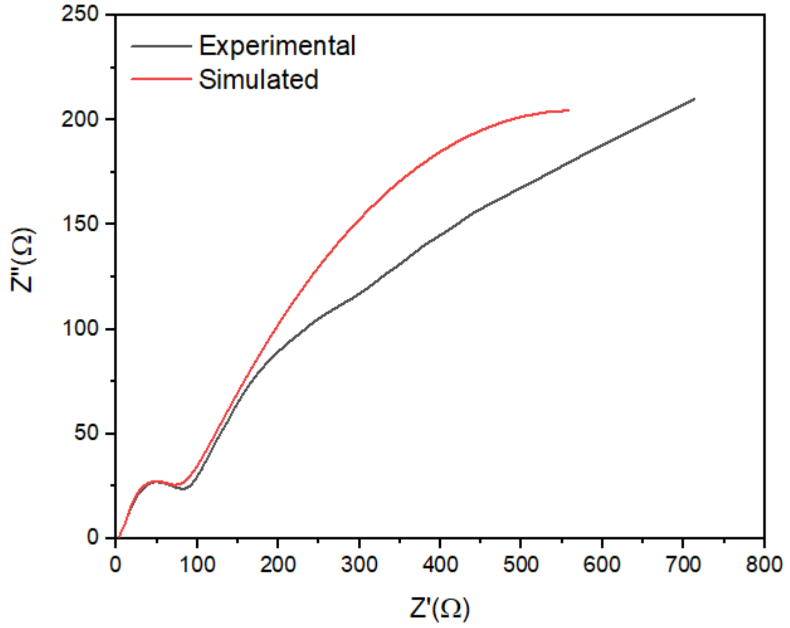
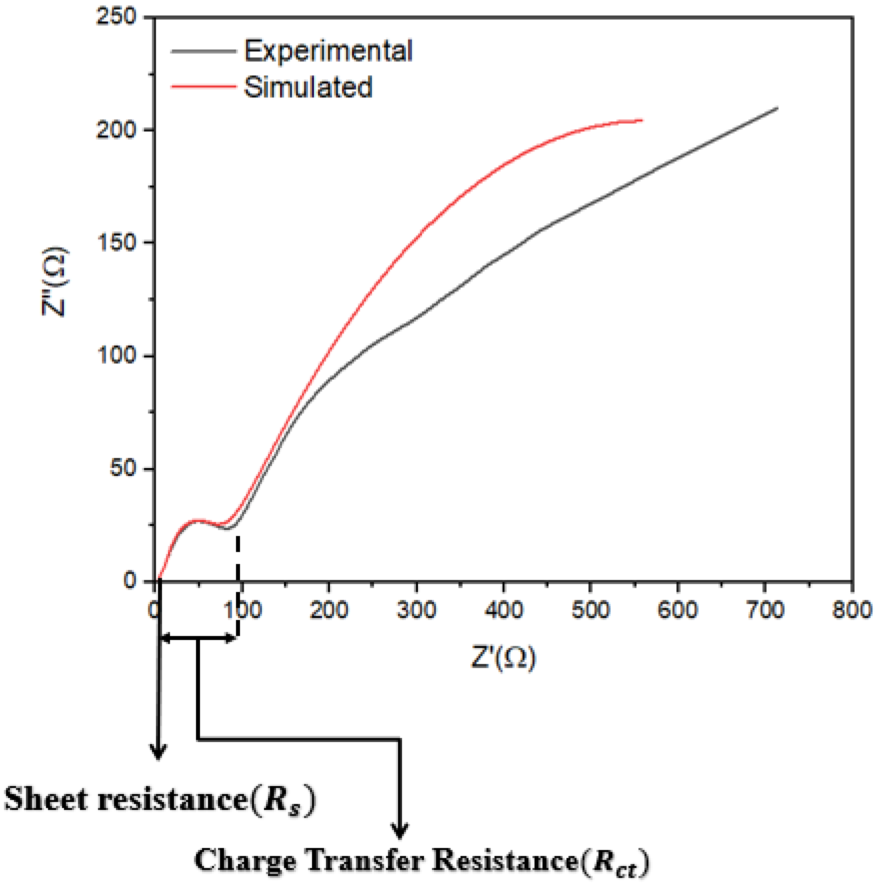


Figure 14 Nyquist plot for real part and imaginary part of Impedance



In the two-electrode EIS measurement, the Nyquist plot was used to evaluate the overall impedance of the complete supercapacitor device. The high-frequency intercept indicates the combined solution resistance (Rs) contributed by the electrolyte, separator, and both electrodes. The semicircle observed at medium frequencies represents the total charge-transfer resistance (Rct) of the full cell, which includes contributions from both positive and negative electrodes. The low-frequency region shows a sloped line corresponding to Warburg impedance (W), highlighting the ion-diffusion limitations through the separator and electrode pores. This analysis provides realistic information about device-level resistance, energy losses, and ion-transport performance under actual operating conditions. Here Charge Transfer Resistant(R_{ct}) is given as 84.997Ω and Series Resistance is given to be 1.094Ω.

For 3 Electrode System

Cyclic Voltammetry

Cyclic voltammetry in a three-electrode setup is used to accurately study the intrinsic electrochemical behaviour of the active material. The reference electrode provides a stable potential, allowing precise measurement of oxidation and reduction processes. This configuration helps identify redox peaks, reversibility, and reaction kinetics. It also gives a clear understanding of the true capacitive or pseudocapacitive nature of the material without influence from the opposite electrode.

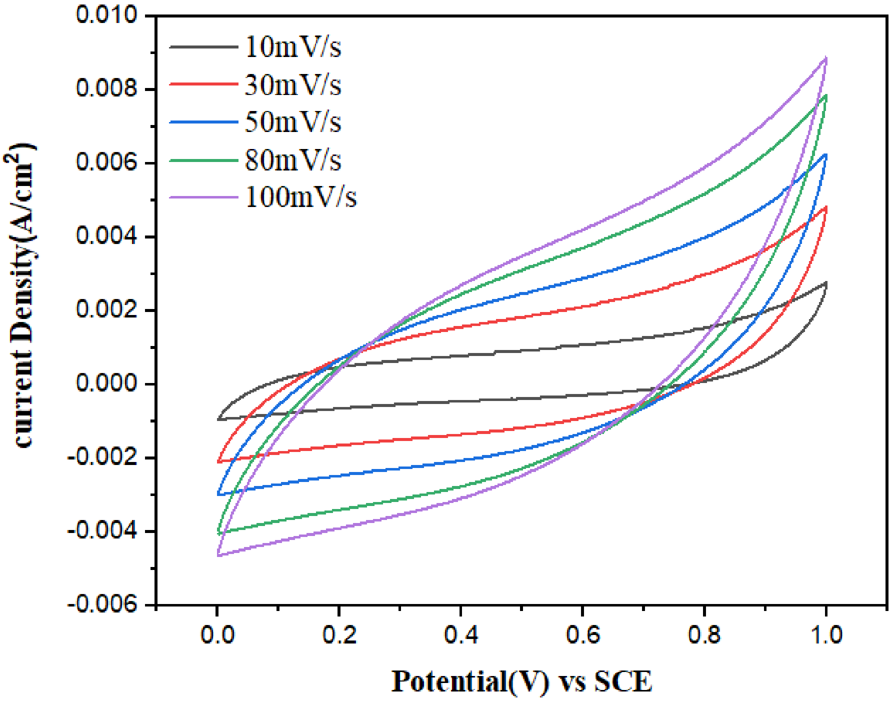


Figure 15 CV graph for 3 electrode system

Calculation of specific Resistance from CV curve:

$$C_s = \frac{Polygon\ area\ from\ Origin}{2Vscan\ rate \times \triangle\ V \times mass\ loaded}$$

$$Units = \frac{Coulombs\ (area)}{2 \times \frac{mV}{s} \times V \times g}$$

Here,

Scan rate=30 $\frac{mV}{s}$ Area=0.0024 g=0.003g

$$C_s = \frac{0.0024}{2 \times 30 \frac{mV}{s} \times 0.5V \times 0.003g}$$

$$C_s = \frac{0.0024}{2 \times (30 \times 10^3) \frac{mV}{s} \times 0.5V \times 0.003g}$$

$$C_s = \frac{0.0024}{2 \times 30 \times 0.5 \times 0.003}\left[\frac{C.s}{g.v^2}\right]$$

$$C_s = \frac{0.0024}{0.00009}\left[\frac{C.s}{g.v^2}\right]$$

$$\because \left[\frac{C.s}{v^2}\right] = F(Farad)$$

$$C_s = \left(\frac{0.0024}{0.00009}\right)\frac{F}{g}$$

$$C_s = 26.66\frac{F}{g}$$

$$\therefore specific\ Capacitance\ from\ Cv\ curve\ is\ calculated\ to\ be\ 26.66\frac{F}{g}$$

Galvanostatic charge discharge(GCD)

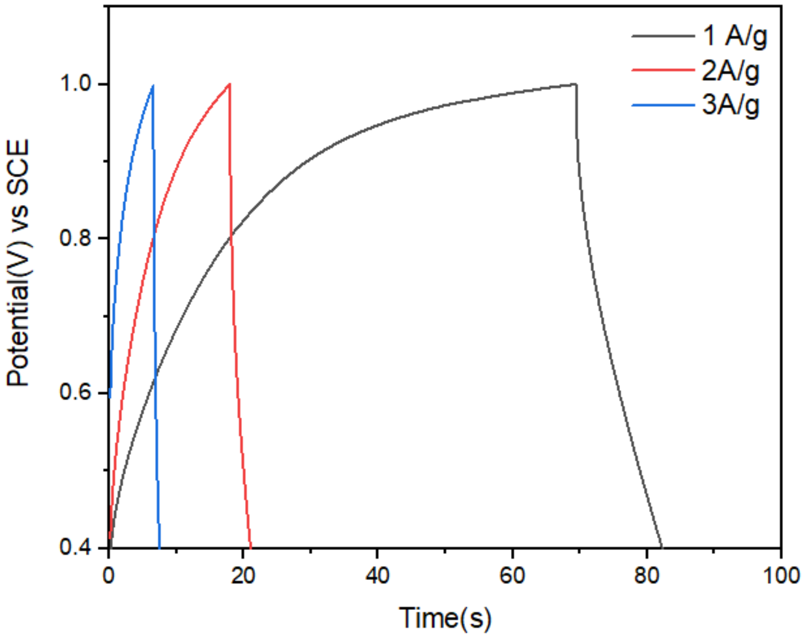
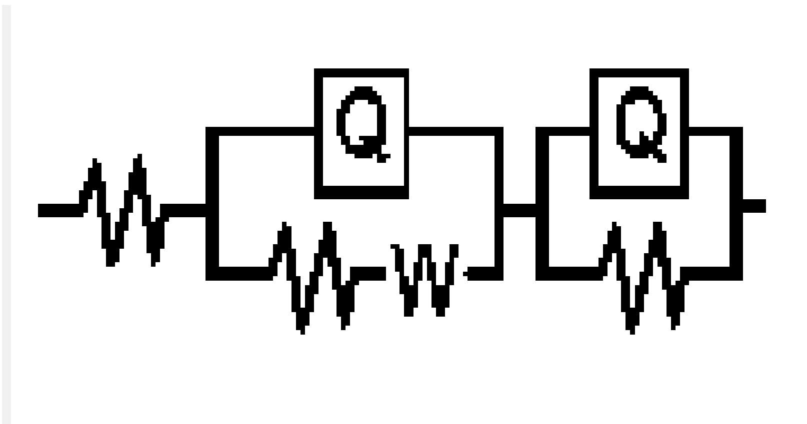


Figure 16 GCD graph for 3 Electrode system

EIS
Circuit for the following Nyquist plot



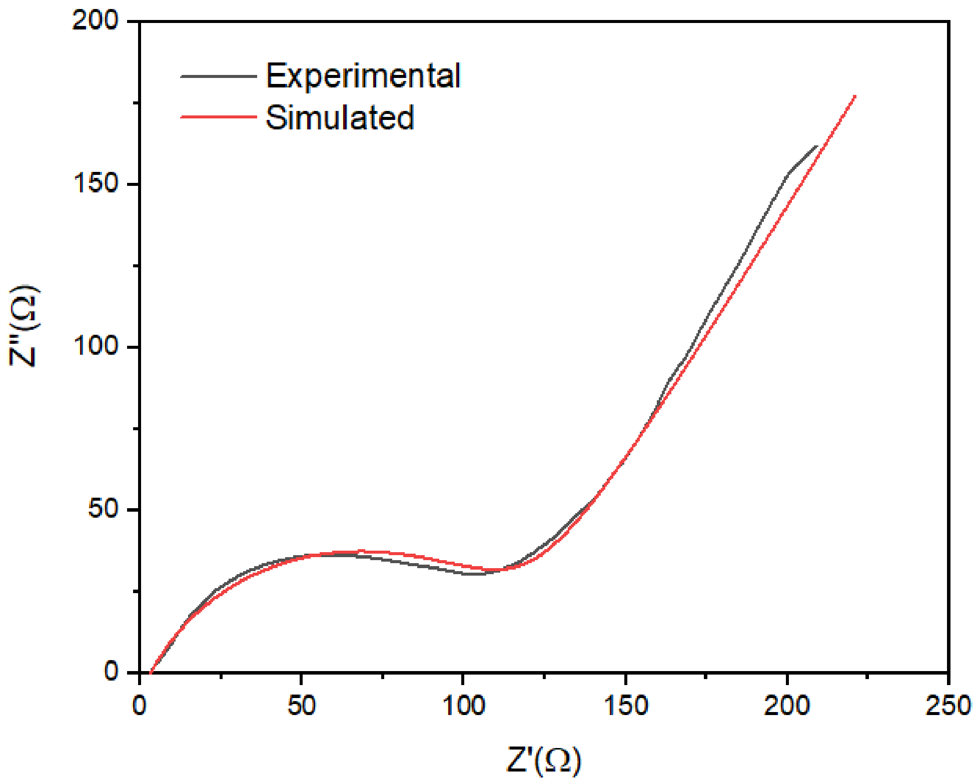
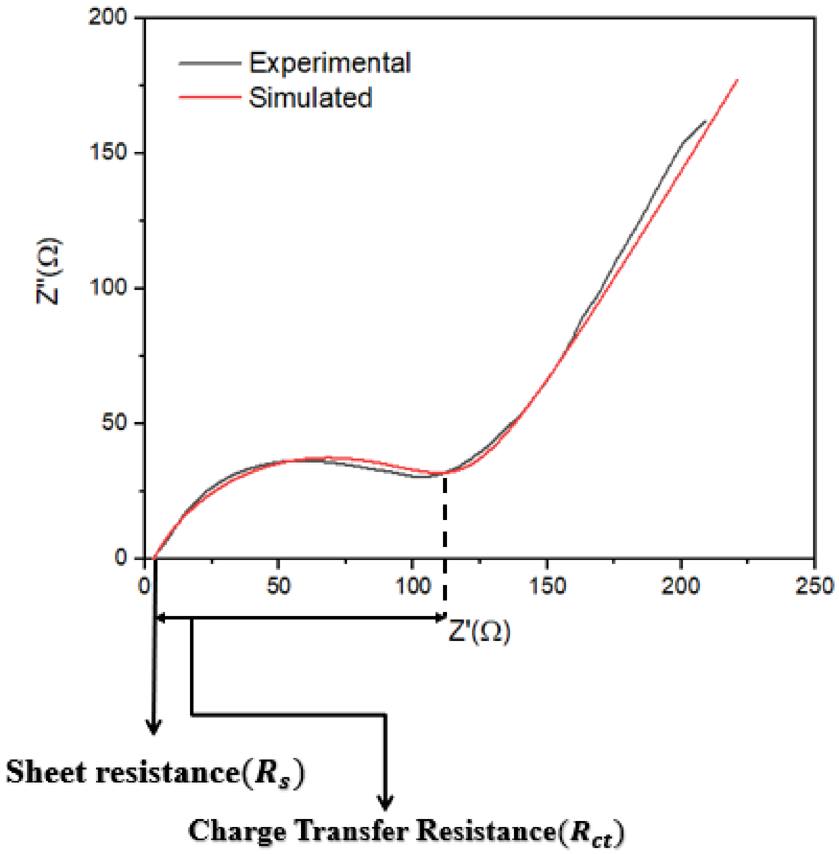


Figure 17 Nyquist plot for real part and imaginary part of Impedance



In the three-electrode EIS analysis, the Nyquist plot was used to study the intrinsic electrochemical behaviour of the active material. The high-frequency intercept on the x-axis represents the solution resistance (Rs), mainly arising from the electrolyte and cell connections. The semicircle in the mid-frequency region corresponds to the charge-transfer resistance (Rct), which reflects the electron-transfer kinetics at the working electrode–electrolyte interface. At lower frequencies, the plot transitions into a slanted line known as Warburg impedance (W), indicating ion diffusion within the porous structure of the material. This setup allows accurate evaluation of the material's conductivity, interfacial properties, and ion-transport behaviour without interference from the counter electrode. Here Charge Transfer Resistant(R_{ct}) is given as 112.85Ω and Series Resistance is given to be 3.248Ω.

ADVANTAGES OF SUPERCAPACITORS

Supercapacitors offer several significant advantages that make them highly suitable for modern energy-storage applications:

1. **High Power Density**
   They can deliver and absorb energy extremely quickly, making them ideal for applications requiring rapid charge–discharge cycles

2. **Long Cycle Life**
   Supercapacitors can endure hundreds of thousands to millions of charge–discharge cycles without significant degradation, far surpassing conventional batteries.

3. **Fast Charging Capability**
   They can be charged within seconds to minutes, providing quick readiness for use.

4. **Wide Operating Temperature Range**
   Supercapacitors perform reliably across a broad range of temperatures, enhancing their usefulness in harsh environments.

5. **High Efficiency**
   They have very low internal resistance, resulting in excellent energy efficiency (typically above 95%) during charging and discharging.

6. **Environmentally Friendly Materials**
   Using biomass-derived activated carbon reduces environmental impact, promotes sustainability, and lowers material cost.

7. **Low Maintenance and High Safety**
   They have lower risk of thermal runaway, minimal maintenance requirements, and better operational stability compared to batteries.

8. **Lightweight and Simple Design**
   Supercapacitors have a straightforward structure, enabling easier fabrication, integration, and scaling for different applications.

LIMITATIONS OF SUPERCAPACITORS

Despite their advantages, supercapacitors also have certain limitations that affect their use in large-scale and long-duration energy-storage applications:

1. **Lower Energy Density Compared to Batteries**
   Supercapacitors store much less energy per unit mass or volume than lithium-ion batteries, making them unsuitable for applications requiring long-term energy supply.

2. **High Self-Discharge Rate**
   They tend to lose stored charge quickly when not in use, which limits their effectiveness in devices that need to hold energy for long periods.

3. **Limited Operating Voltage**
   Most supercapacitors operate at low voltage ranges, and stacking multiple cells increases cost, complexity, and balancing requirements.

4. **Higher Cost for High-Performance Materials**
   Although biomass-derived carbon can reduce costs, advanced materials like metal oxides or conducting polymers can be expensive to produce and scale up.

5. **Temperature Sensitivity**
   Their performance can degrade at very high or very low temperatures, affecting reliability in extreme environmental conditions.

6. **Bulky Design for High Energy Applications**
   To match the energy of a battery, a supercapacitor device would need to be much larger and heavier, limiting its use in compact systems.

7. **Limited Long-Term Energy Storage**
   They are ideal for fast charge–discharge cycles but not for applications where energy must be stored for hours or days.

APPLICATIONS OF SUPERCAPACITORS

Supercapacitors have emerged as promising energy-storage devices due to their high-power density, long cycle life, and rapid charge–discharge capability. The materials developed and tested in this project—biomass-derived activated carbon electrodes—further enhance their sustainability and cost-effectiveness. Key applications include:

1. **Electric Vehicles (EVs) and Hybrids**
   Supercapacitors provide fast acceleration power, regenerative braking energy capture, and support for sudden load demands, improving overall vehicle efficiency and battery lifespan.

2. **Backup Power Systems**
   They are widely used in UPS systems, data centres, and industrial electronics to provide immediate backup power and smooth transitions during power interruptions.

3. **Renewable Energy Storage**
   Supercapacitors help stabilize energy fluctuations in solar and wind systems by quickly absorbing or supplying power, improving grid reliability.

4. **Portable and Consumer Electronics**
   Applications include power stabilization in mobile devices, wearables, cameras, and smart sensors due to their fast response and long operational life.

5. **Memory Backup and CMOS Systems**
   They are used for maintaining memory, clock functions, and data protection in electronics during sudden power loss.

6. **Grid-Level and Microgrid Applications**
   Supercapacitors support load levelling, frequency regulation, and peak-power management in smart-grid infrastructure.

7. **Industrial and Transportation Systems**
   They are implemented in cranes, metros, buses, and heavy machinery for energy recovery, peak-load handling, and improving system efficiency.

8. **Internet of Things (IoT) Devices**
   Supercapacitors provide reliable power for sensors and autonomous devices, especially in environments requiring long cycle life and rapid recharge.

PURPOSE OF THE STUDY

The primary purpose of this study was to develop and evaluate sustainable, high-performance electrode materials for supercapacitors using biomass-derived activated carbon. The work aimed to understand how eco-friendly carbon sources—such as paddy waste and bamboo powder—can be converted into efficient energy-storage materials through activation, carbonization, and electrode fabrication. Additionally, the study sought to compare the electrochemical behaviour of the synthesized materials using both two-electrode and three-electrode systems through CV, GCD, and EIS techniques. This dual analysis helped assess both intrinsic material properties and practical device performance. Overall, the study was designed to contribute to the development of low-cost, environmentally friendly supercapacitor technologies with potential applications in renewable energy, electric mobility, and future sustainable power systems.

CONCLUSION

The internship at IIT Madras provided valuable hands-on experience in the fabrication and electrochemical analysis of biomass-derived supercapacitor materials. Through systematic synthesis, activation, and carbonization of bio-waste precursors, followed by electrode preparation and device assembly, I gained practical insights into sustainable material development for energy-storage applications. The comparative study of two-electrode and three-electrode systems using CV, GCD, and EIS techniques helped me understand both intrinsic material properties and real-device performance. The results demonstrated the potential of activated carbon–based electrodes for high-performance supercapacitors, highlighting their efficiency, stability, and eco-friendly nature. Overall, this internship strengthened my technical skills, deepened my understanding of electrochemical energy storage, and motivated me to continue exploring research in advanced materials and green technologies.

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